

ALIER RENG

HEALTHCARE QUALITY · SAFETY · COMPLIANCE LEADER

Data-driven director with 9+ years across CMS quality programs, NHSN safety reporting, and clinical operations analytics.

alierwai254@gmail.com

Louisville, KY · open to relocation

[linkedin.com/in/tongakuot](https://www.linkedin.com/in/tongakuot)

github.com/tongakuot

Healthcare quality and analytics leader with **9+ years** partnering with Clinical Operations, Quality, and Care Management leaders to operationalize **CMS quality programs** (VBP, HAC Reduction, MSPB), **NHSN safety reporting**, and the **National Quality Strategy (NQS)** across multi-hospital systems. Recognized for rigorous data stewardship and translating complex measure specifications into actionable scorecards that improve quality and patient safety. LifePoint Health **Mercy Award** recipient (2020).

PROFESSIONAL EXPERIENCE

Director, Analytics & Quality Reporting · ScionHealth

2022 — Present

Louisville, KY

- ▶ Lead data stewardship for quality and operational reporting across the **Community Hospitals Division** — designing initial and monthly data-validation protocols that protect the integrity of quality and safety metrics.
- ▶ Partner with SMEs and Clinical Operations leaders to develop, document, and operationalize **National Quality Strategy (NQS)** measures — including specifications, calculation logic, and audit trails.
- ▶ Author **Quality Measures Specifications** and field-level reporting instructions used by hospital quality and compliance leaders.
- ▶ Compile and compress monthly **NQS Excel workbooks for 75+ hospitals enterprise-wide** with R and openpyxl — cutting production time from **several days to 4 hours** while improving accuracy and consistency.
- ▶ Automate divisional Excel reporting with **Python (openpyxl)** — the **Apogee and TeamHealth GMLOS Variance** reports and the **Quality & Regulatory Outcomes HCAHPS and ED CAHPS trend** reports — strengthening accuracy and audit-readiness across patient-experience, safety, and utilization metrics.
- ▶ Build **R Shiny** applications for monthly Operating Review (MOR) data — giving executive and frontline leaders interactive access to quality, safety, and throughput indicators.

Senior Data Analyst, Quality Data · LifePoint Health

2021 — 2022

Brentwood, TN

- ▶ Data stewardship: performed monthly data validation to ensure data accuracy before data was pushed to PowerBI for hospitals' use.
- ▶ Supported and assisted hospitals and the HSC departments with their data needs.
- ▶ Performed exploratory and predictive analysis with R and Python.

QUALITY & COMPLIANCE

- CMS VBP · HAC Reduction · MSPB
- HCAHPS & ED CAHPS trend reporting
- NHSN safety & infection reporting
- National Quality Strategy (NQS)
- Quality Measures Specifications
- QOC, MOR, GMLOS variance
- Mortality (CHOIS) reporting
- Data stewardship & validation
- Audit-readiness & documentation

ANALYTICS & TOOLS

- R · Python · SQL
- R Shiny · Shiny for Python
- openpyxl · Quarto · tidymodels
- PowerBI dashboards & DAX
- pandas · Polars · scikit-learn
- Positron · RStudio · VS Code
- JupyterLab · Git / GitHub · MS Office

LEADERSHIP STRENGTHS

- Cross-functional partnership with Clinical Ops & Care Management
- Translating regulatory measures into operational dashboards
- Coaching, mentorship, and team development
- Executive-ready written communication

EDUCATION

MBA · Information Systems Management

2016 — 2017

LeTourneau University · Longview, TX

- ▶ Created and modified divisions' and hospitals' **NHSN Goals, MOR, and QOC** reports and dashboards using R and PowerBI.
- ▶ Onboarded and trained new team members.

Data Analyst, Quality Data · LifePoint Health 2016 — 2021
Brentwood, TN

- ▶ Cut **QOC and MOR** decks production time by **80%** by automating reports with R Shiny.
- ▶ Reduced data errors and optimized accuracy by automating **CHOIS monthly mortality** reports with R Shiny.
- ▶ Used R to automate the CMS company-wide **VBP, HAC Penalty, and MSPB** reports extraction processes.

TEACHING & PUBLIC EDUCATION

Lecturer, Mathematics · Middle Tennessee State University 2011 — 2016
Murfreesboro, TN

- ▶ Taught general mathematics courses — Plane Trigonometry, Applied Calculus I, Applied Statistics, Finite Mathematics, College Algebra, Mathematics for General Studies (Mathematical Ideas), and Pre-Calculus (Pathways to Calculus).
- ▶ Turned 'F' students into 'A' and 'B' students through active and participatory learning; used hands-on approaches to help students conceptualize mathematical ideas and construct meaningful mathematical statements.
- ▶ Helped challenged students replace negative thinking with positive-possibility mindsets through friendly coaching and mentorship.

LEADERSHIP & COMMUNITY

Founder & Lead Educator · PyStatR+ — mission-driven data science education 2024 — Present
Louisville, KY

- ▶ Founded the first South Sudanese-American, U.S.-based data science education initiative — delivering accessible education in data science, statistics, ML, and AI with a focused commitment to learners in South Sudan and East Africa.
- ▶ Author open tutorials in R, Python, and statistics; publish across the PyStatR+ YouTube channel, Facebook community, Medium, and blog with reproducible Quarto notebooks.
- ▶ Design and deliver group training programs (1-day, 5-day, 8-week) for students, professionals, NGOs, and government agencies — sharpening curriculum design, reproducibility, and technical communication skills.

M.S. · Professional Science (Biostatistics)

2009 — 2011
Middle Tennessee State University · Murfreesboro, TN

B.S. · Neuroscience

2005 — 2008
University of Texas at Dallas · Richardson, TX

AWARDS

Mercy Award · 2020

LifePoint Health — quality mission contribution

LSAMP Scholarship · 2006-2007

The University of Texas at Dallas

Vice President's List · 2005

Richland College (DCCCD)